

KHAGENDRA NEUPANE

Aspiring AI/ML Engineer & Researcher | B.E. Electronics, Communication & Information Engineering

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RESEARCH INTERESTS & OBJECTIVE

Motivated undergraduate engineering student seeking to pursue advanced study in Artificial Intelligence and Machine Learning. Background in systems and network programming provides a strong computational foundation, now applied toward building intelligent, AI-driven software systems. Particularly interested in applied machine learning, AI-powered automation, and the intersection of AI with secure, resilient systems design. Seeking a graduate program / scholarship that will provide rigorous research training and mentorship in ML theory and applications.

EDUCATION

Tribhuvan University — Pulchowk Campus (Institute of Engineering)

Expected 2028

Bachelor of Engineering, Electronics, Communication & Information Engineering (BEI)

- Coursework and self-study directed toward AI/ML, data structures & algorithms, and computer networks.
- Elected Class Representative, Batch BEI081.

Urbana School of Science

Graduated 2024

+2 Science • GPA: 3.92 / 4.0

LEADERSHIP & AWARDS

President, Ignite Open Source Community

Present

- Lead a student-run open-source community, setting direction and overseeing programs that introduce members to collaborative software development.
- Organize workshops, contribution sprints, and mentorship sessions to grow members' skills in Git/GitHub, Python, and applied AI/ML tooling.
- Coordinate a core team of contributors and maintain partnerships with peer tech communities.

Winner (Team), Hack A Week 2026

2026

- Won a week-long team hackathon, delivering a working software prototype under time constraints alongside a cross-functional team.
- Contributed to system design, backend implementation, and integration of AI-assisted features into the winning solution.

Open Source Contributor

Ongoing

- Active contributor to community open-source projects on GitHub: writing code, improving documentation, and resolving issues in collaboration with maintainers worldwide.

SELECTED PROJECTS

AI Desktop Assistant

- Voice-controlled desktop assistant that uses AI/NLP to interpret natural-language commands, answer questions, and perform actions such as sending emails and WhatsApp messages.
- Integrated speech recognition with an AI response-generation pipeline to provide conversational, hands-free assistance.

Setu

- An AI-powered platform for legal assistance in Nepal – making legal documents accessible, generating official letters, and detecting bias in legal text.

kn-sock — Python Networking Toolkit

- Designed and published a lightweight, Pythonic socket-programming library that simplifies building networked applications.

Other Systems Projects

- ArchShell — a custom command shell for Linux-based systems.
- P2P Chat Application — a peer-to-peer chat system built on raw sockets.
- Lightweight HTTP Server — implemented from first principles using Python sockets.

TECHNICAL SKILLS

Programming: Python (Intermediate), C/C++ (Advanced), Java (Basics)
AI/ML & Applied Interests: AI-driven automation, NLP-based assistants, AI/ML applications in software and cyber defense
Backend & Web: Flask, FastAPI, HTML/CSS, socket programming
Systems & Tools: Linux CLI (Ubuntu, Arch), Git & GitHub, shell scripting

LANGUAGES

Nepali (Native) • English (Fluent) • Hindi (Conversational)